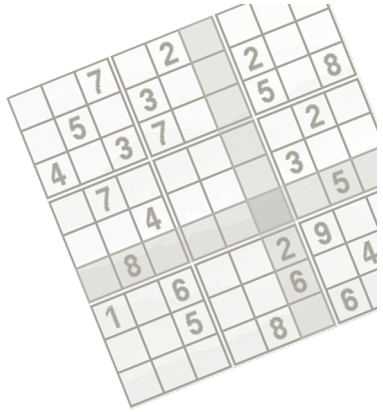

Constraint Programming

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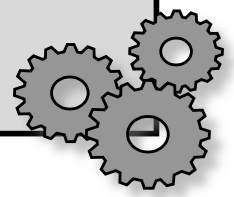
Look-back Search Algorithms

Back to **generate-and-test**:

- **generates** a solution candidate (a complete assignment) and **tests** all the constraints together at the end
- solution candidates are generated systematically, for example:

```
procedure generate_first(Variables)
        for each V in Variables do
                label V by the first value in D_V
        end for
end generate_first

procedure generate_next(Assignment)
        find first X in Assignment such that all following variables are labelled by the last value
        from their respective domains (name the set of these variables Vs)
        if X is labelled by the last value then return fail
        label X by next value in D_X
        for each Y in Vs do
                assign first value in D_Y to Y
        end for
end generate_next
```



We can verify satisfaction of a constraint as soon as we know the values of all constrained variables!

- the test stage is done during the generation stage

↳ tzn můžu přestat generovat, když už stejně nesplňují podmínky

Probably the most widely used systematic search algorithm that **verifies the constraints as soon as possible**.

- upon failure (any constraint is violated) the algorithm goes back to the last instantiated variable and tries a different value for it
- depth-first search

The core principle of applying backtracking to solve a CSP:

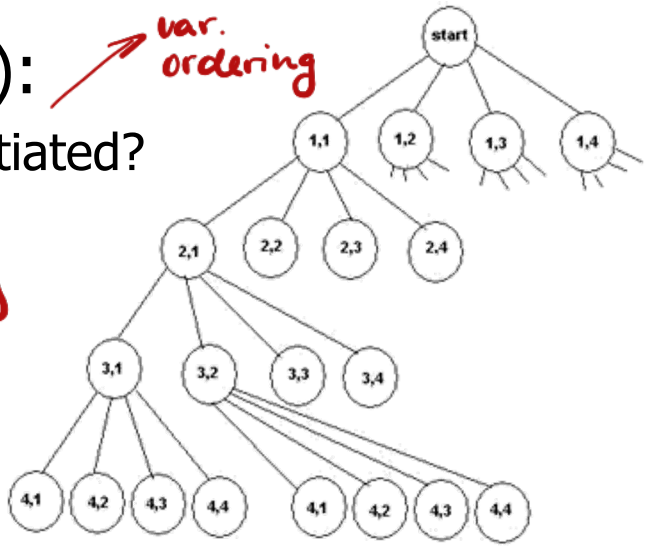
1. assign values to variables one by one
2. after each assignment verify satisfaction of constraints with known values of all constrained variables

Open questions (to be answered later):

- What is the order of variables being instantiated?
- What is the order of values tried?

var. ordering

val. ordering

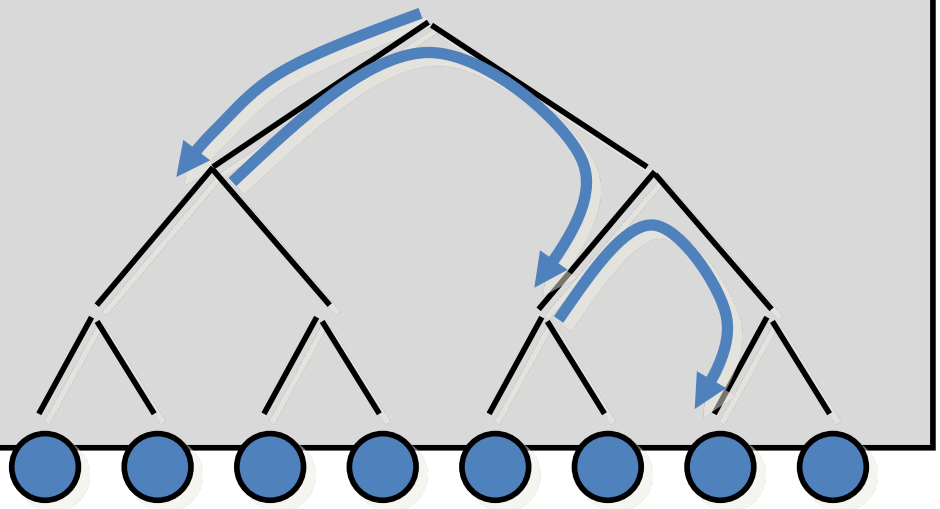


Backtracking explores partial consistent assignments until it finds a complete (consistent) assignment.

Chronological Backtracking (a recursive version)

```
procedure BT(X:variables, V:assignment, C:constraints)
    if X={} then return V
    x ← select a not-yet assigned variable from X
    for each value h from the domain of x do
        if constraints C are consistent with V ∪ {x/h} then
            R ← BT(X – {x}, V ∪ {x/h}, C)
            if R ≠ fail then return R
    end for
    return fail
end BT
```

Call as BT(X, {}, C)



Note:

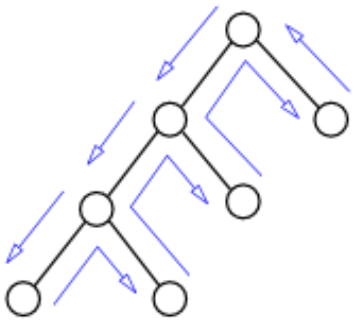
If it is possible to perform the test stage for a partially generated solution candidate then BT is always better than GT, as BT does not explore all complete solution candidates.

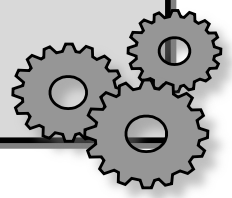
Chronological Backtracking (an iterative version)

```
procedure BT(X:variables, C:constraints)
    i ← 1, D'_i ← D_i
    while 1 ≤ i ≤ n do
        instantiate_and_check(i, C)
        if x_i = null then i ← i – 1 else i ← i + 1, D'_i ← D_i
    end while
    if i = 0 then return fail
    return {x_1,..., x_n}
end BT
```

vracím se pokračuji

```
procedure instantiate_and_check(i, C:constraints)
    while D'_i is not empty do
        select and delete some element b from D'_i
        x_i ← b
        if constraints C are consistent with {x_1,..., x_i} then return
    end while
    x_i ← null
end instantiate_and_check
```





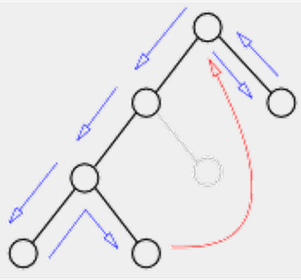
Weaknesses of Backtracking

Look Back

- **thrashing**
  - throws away the reason of the conflict

  Example: A,B,C,D,E:: 1..10, A>E
    - BT tries all the assignments for B,C,D before finding that A≠1

  Solution: **backjumping** (jump to the source of the failure)



- **redundant work**
  - unnecessary constraint checks are repeated

  Example: A,B,C,D,E:: 1..10, B+8<D, C=5*E
    - when labelling C,E the values 1,..,9 are repeatedly checked for D

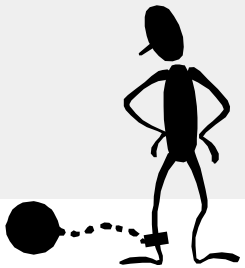
  Solution: **backmarking, backchecking** (remember (no-)good assignments)

Look Ahead

- **late detection of the conflict**
  - constraint violation is discovered only when the values are known

  Example: A,B,C,D,E::1..10, A=3*E
    - the fact that A>2 is discovered when labelling E

  Solution: **forward checking** (forward check of constraints)



Backjumping is a method to **remove thrashing.**

How to do it?

1) identify the source of the conflict (impossibility to assign a value)
2) jump to the past variable in conflict

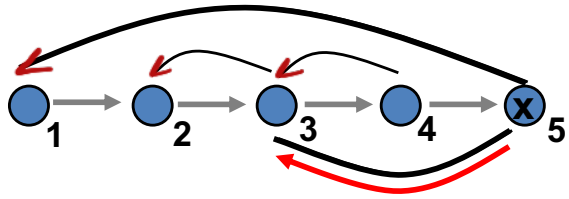
The same forward run as in backtracking, only the back-jump can be longer to skip irrelevant assignments!

How to find a jump position? What is the source of the conflict?

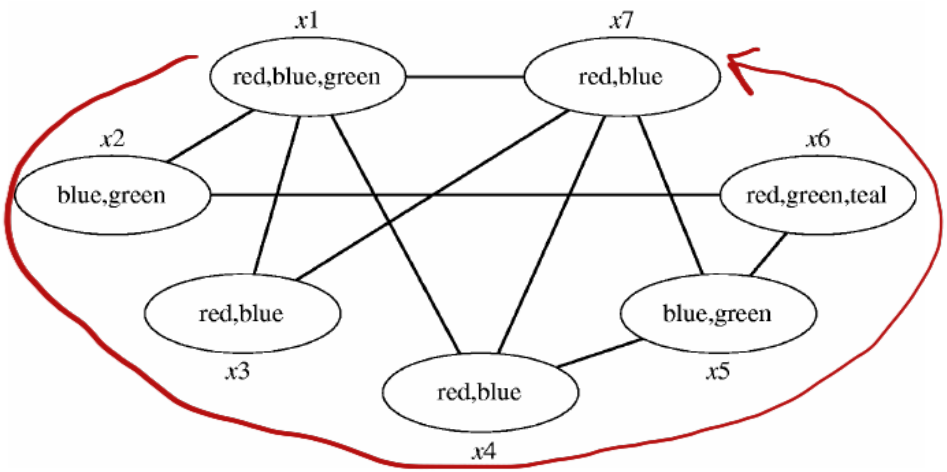
- select the constraints containing just the currently instantiated variable and the past variables
- select the closest variable participating in the selected constraints

Graph-directed backjumping

nevím, co selhalo → jdu do nejbližší propojené

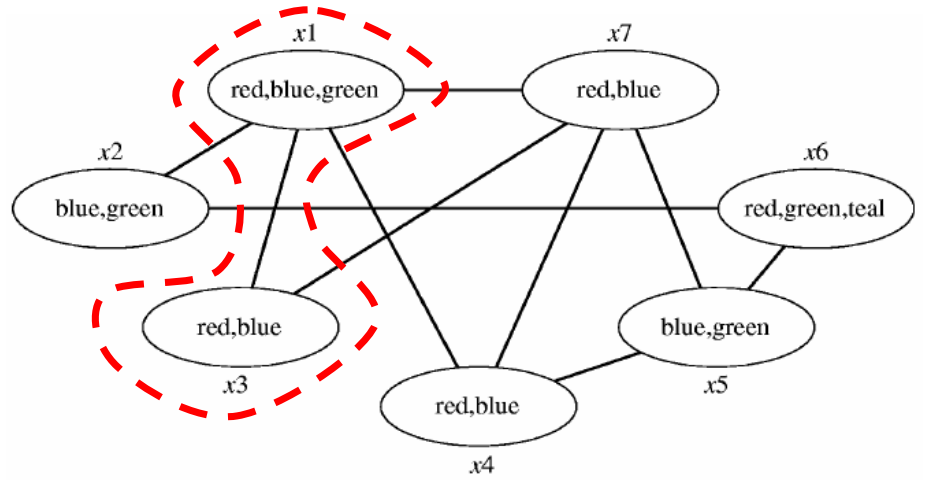


Assume the **graph colouring** problem, where the nodes are coloured in the order x1, x2, …, x7.



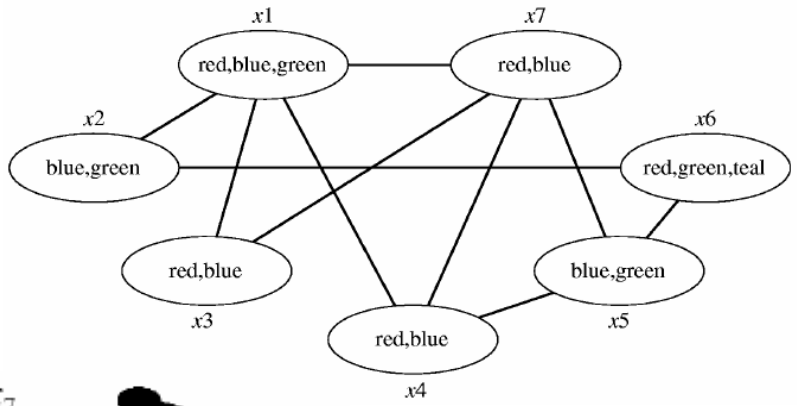
- Where to jump if colouring x4 fails?
  - to **x1**
- Where to jump if colouring x5 fails?
  - to **x4**
    - And what if x4 cannot be coloured?
  - to **x1** 5 nemá jinou hranu, tak koukám na 4

It looks like after failure with some node, we can **jump to its closest predecessor** (in the colouring order), but …

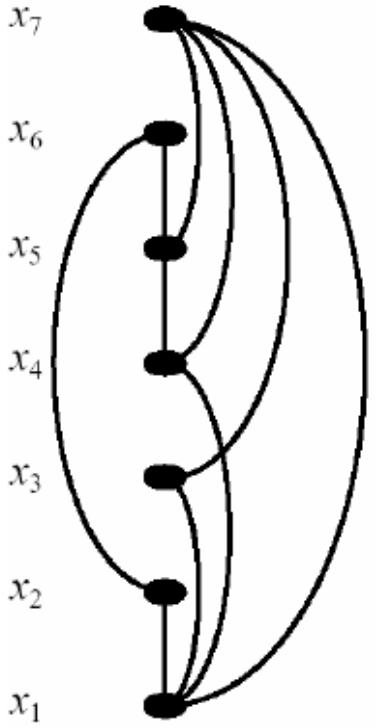


- Where to jump if colouring x7 fails?
- to **x5**
  - What if colouring x5 fails now?
  - to **x4** koukám ze 7
    - And what if x4 cannot be coloured?
  - to **x3** opět koukám ze 7

When jumping back, it is enough to free some **dead-end** (dead-end = a node/variable, that cannot be instantiated).



- from the leaf x jump to the closest predecessor of x in the constraint network
  (x7 → x5, x4, x3, x1)
- from the inner node x jump to the closest predecessor of all dead-end nodes visited during the jumps
  (x7 → x5, x4, x3, x1 → x4, x3, x1 → x3, x1)



– let **anc(x)** be the **ancestors of x** in the constraint network ordered using the labelling order (can be decided based on the network structure)
  - anc(x7) = {x5, x4, x3, x1}

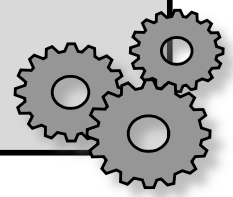
jump set

– let us backjump to node x from the nodes y1,…,yk and let there be no more value for variable x
– then **jump to the closest variable from the set anc(x) ∪ anc(y1) ∪ … ∪ anc(yk) – {x,y1,…,yk}**

Graph-Directed Backjumping (a recursive version)

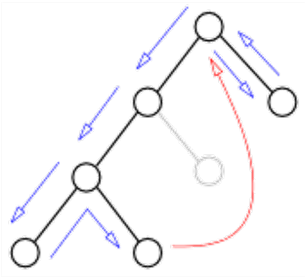
```
procedure GraphBJ(X:variables, V:assignment, C:constraints)
    if X = {} then return V
    x ← select a not-yet assigned variable from X
    conflict ← anc(x)
    for each value h from the domain of x do
        if constraints C are consistent with V ∪ {x/h} then
            R ← GraphBJ(X – {x}, V ∪ {x/h}, C)
            if R = fail(JumpSet) then                  % backjump
                if x∉JumpSet then return R             % to a variable before x
                conflict ← conflict ∪ JumpSet – {x}    % to x
            else return R                              % solution found
    end for
    return fail(conflict)
end GraphBJ
```

Call as GraphBJ(X, {}, C)

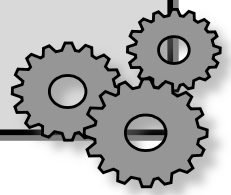


Graph-Directed Backjumping (an iterative version)

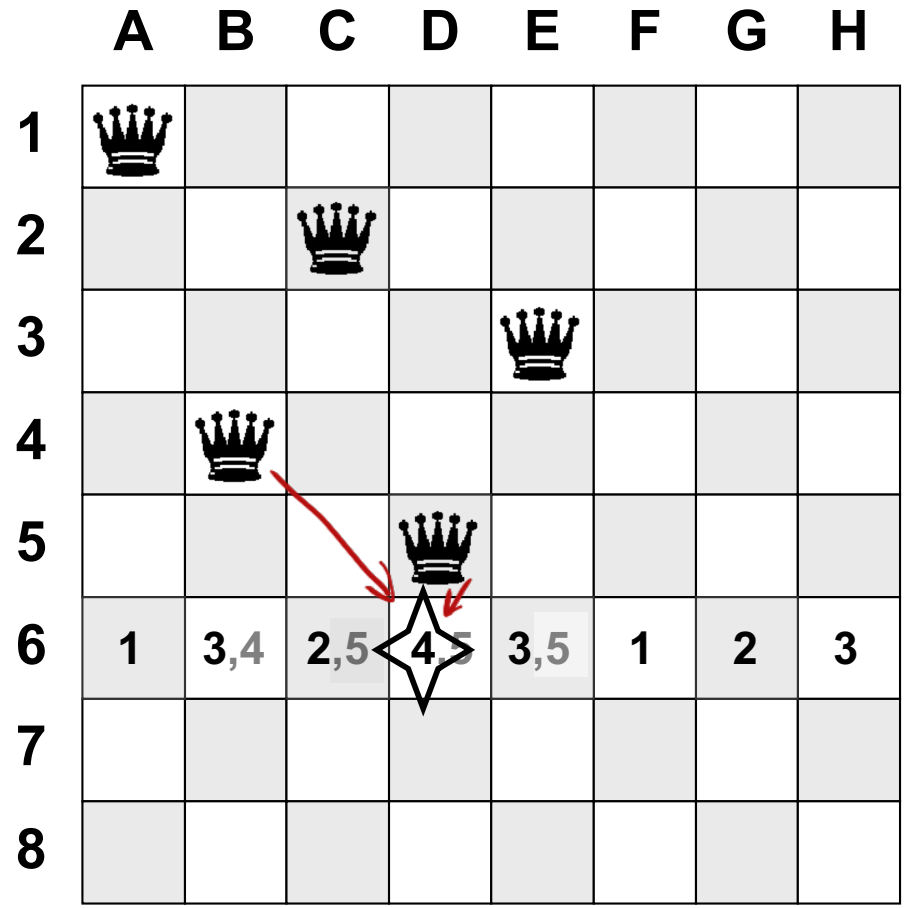
```
procedure GraphBJ(X:variables, C:constraints)
    i ← 1, D'_i ← D_i , l_i ← anc(x_i)
    while 1 ≤ i ≤ n do
        instantiate_and_check(i, C)
        if x_i = null then
          iprev ← i, i ← latest index in l_i, l_i ← l_i ∪ l_iprev − {x_i}
        else
          i ← i + 1, D'_i ← D_i , l_i ← anc(x_i)
        end if
    end while
    if i = 0 then return fail
    return {x_1,..., x_n}
end GraphBJ
```



```
procedure instantiate_and_check(i, C:constraints)
    while D'_i is not empty do
        select and delete some element b from D'_i
        x_i ← b
        if constraints C consistent with {x_1,..., x_i} then return
    end while
    x_i ← null
end instantiate_and_check
```



N-queens problem



Queens in rows are allocated to columns.

6th queen cannot be allocated!

1. Write a number of conflicting queens to each position.

2. Select the farthest conflicting queen for each position.

3. Select the closest conflicting queen among positions.

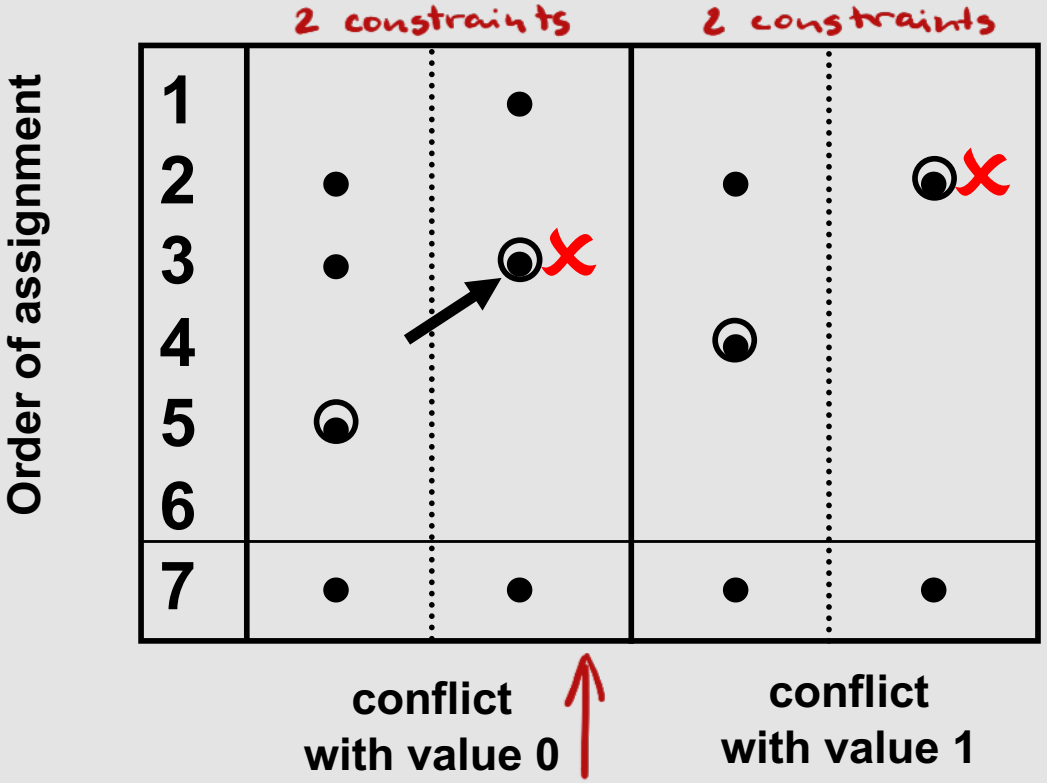
Note:

Graph-directed backjumping has no effect here (due to a complete graph)!

How to find out the conflicting variable?

Situation:

- assume that the variable no. 7 is being assigned (values are 0, 1)
- the symbol • marks the variables participating in the violated constraints (two constraints for each value)



Neither 0 nor 1 can be assigned to the seventh variable!

1. Find the closest variable in each violated constraint (o).

2. Select the farthest variable among the above chosen variables for each value (×).

3. Choose the closest variable among the conflicting variables selected for each value and jump to it.

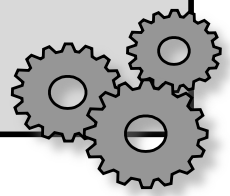
Gaschnig Backjumping – consistency check

In addition to consistency check we can also find out the conflicting level!

```
procedure consistent(Labelled, Constraints, Level)
    J ← Level                    % the level to jump to
    NoConflict ← true            % is there any conflict?
    for each C in Constraints do
      if all variables from C are Labelled then
          if C is not satisfied by Labelled then
                NoConflict ← false
                J ← min {J, max{L | X ∈ vars(C) & X/V/L in Labelled & L<Level}}
          end if
      end if
    end for
    if NoConflict then return true
                  else return fail(J)
end consistent
```

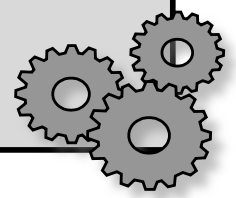
calculating jump level

- **V is a value of X**
- **L is the depth level of X during labelling**



Gaschnig Backjumping (a recursive version)

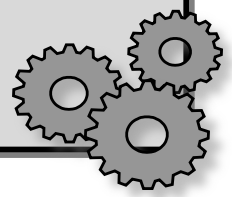
```
procedure GBJ(Unlabelled, Labelled, Constraints, PreviousLevel)
    if Unlabelled = {} then return Labelled
    pick first X from Unlabelled
    Level ← PreviousLevel+1
    Jump ← 0
    for each value V from D_X do
      C ← consistent({X/V/Level} ∪ Labelled, Constraints, Level)
      if C = fail(J) then
          Jump ← max {Jump, J}
      else
          Jump ← PreviousLevel
          R ← GBJ(Unlabelled-{X},{X/V/Level} ∪ Labelled,Constraints, Level)
          if R ≠ fail(Level) then return R        % success or jump further
      end if
    end for
    return fail(Jump)                             % jump to the conflicting variable
end GBJ

Call as GBJ(Variables,{},Constraints,0)
```



Gaschnig Backjumping (an iterative version)

```
procedure GBJ(X:variables, C:constraints)
    i ← 1, D'_i ← D_i , jump_i ← 0
    while 1 ≤ i ≤ n do
        x_i ← select_value(i, C)
        if x_i = null then
            i ← jump_i
        else
            i ← i + 1
            D'_i ← D_i
            jump_i ← 0
        end if
    end while
    if i = 0 then return fail
    return {x_1,..., x_n}
end GBJ
```

```
procedure select_value(i, C:constraints)
    while D'_i is not empty do
        select and delete some element b from D'_i
        consistent ← true
        k ← 1
        while k<i and consistent do
            if k>jump_i then jump_i ← k
            if x_i=b consistent with {x_1,..., x_k} in C then
                    k ← k + 1
            else    consistent ← false
        end while
        if consistent then return b
    end while
    return null
end select_value
```



Backjumping – a short summary

- **Graph-directed Backjumping**
  - driven by the structure of constraint network only (does not assume (dis)satisfaction of constraints)
  - can do several jumps in a sequence

pro nQ je na nic
→ složitější než BT
a nedělá nic
lepšího pro nQ

- **Gaschnig Backjumping** → skáče jen z listu
  - assumes which constraints are violated
  - just one back-jump (if the assignment fails again, only one level up is backtracked like in chronological backtracking)

nQ ✓

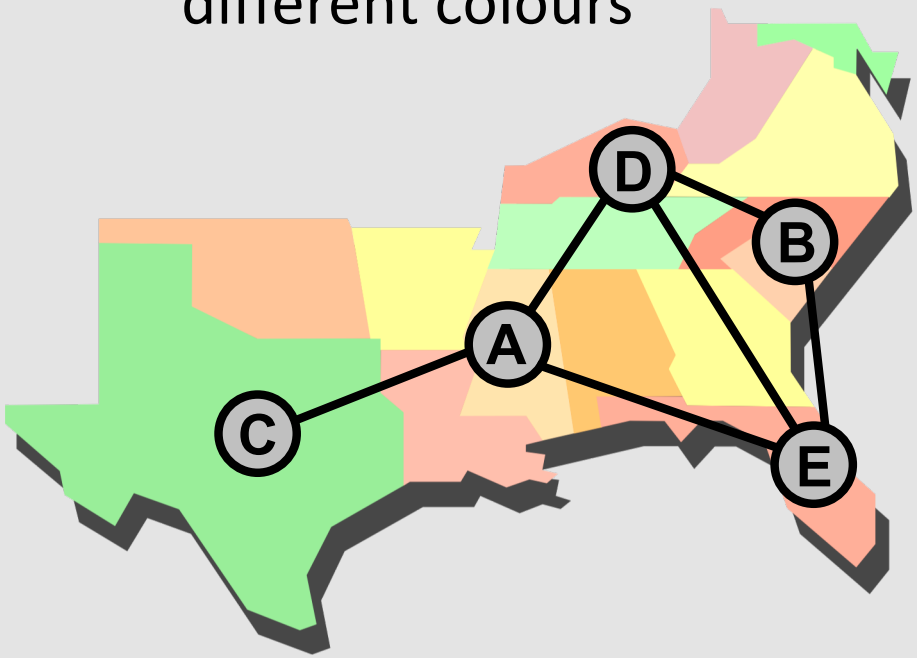
- **Conflict-driven Backjumping (CBJ)**
  - we can join advantages of both methods (better target to jump to and more back-jumps in a sequence)
  - when jumping back we need to keep a **conflict set** of variables that is used for the next jump if no value is found for the current variable
    - we carry the source of conflict when backjumping

nQ ✓

When jumping back the in-between assignment is lost!

Example:

– colour the graph in such a way that the connected vertices have different colours

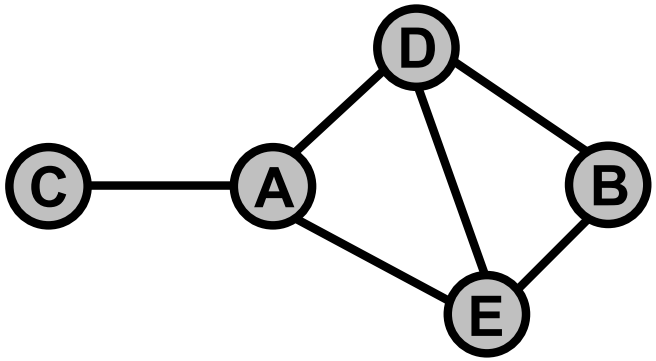


node	vertex		
A	1		1
B	2		1 2
C	1 2	backjump	1 2
D	1 2 3		1 2
E	1 2 3		1 2 3

During the second attempt to label C superfluous work is done - it is enough to leave there the original value 2, the change of B does not influence C.

Dynamic Backtracking in an example

The same graph (A,B,C,D,E), the same colours (1,2,3) but a different approach.



Backjumping
+ remember the source of the conflict
+ carry the source of the conflict
+ change the order of variables

= DYNAMIC BACKTRACKING

node	1	2	3
A	•		
B		•	
C	A	•	
D	A	B	•
E	A	B	D

jump back
+ carry the conflict source

node	1	2	3
A	•		
B		•	
C	A	•	
D	A	B	AB
E	A	B	

jump back
+ carry the conflict source
+ change the order of B, C

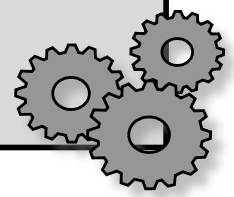
node	1	2	3
A	•		
C	A	•	
B	•	A	
D	A	•	
E	A	D	•

• selected colour
AB a source of the conflict

The vertex C (and the possible sub-graph connected to C) is not re-coloured.

Algorithm Dynamic Backtracking

```
procedure DB(Variables, Constraints)
    Labelled ← {};    Unlabelled ← Variables
    while Unlabelled ≠ {} do
        select X in Unlabelled
        Values_X ←  D_X - {values inconsistent with Labelled using Constraints}
        if Values_X = {}  then
            let E be an explanation of the conflict (set of conflicting variables)
            if E = {}  then failure
            else
                let Y be the most recent variable in E
                unassign Y (from Labelled) with eliminating explanation E-{Y}
                remove all the explanations involving Y
            end if
        else
            select V in Values_X
            Unlabelled ←  Unlabelled - {X}
            Labelled ← Labelled ∪ {X/V}
        end if
    end while
    return Labelled
end DB
```

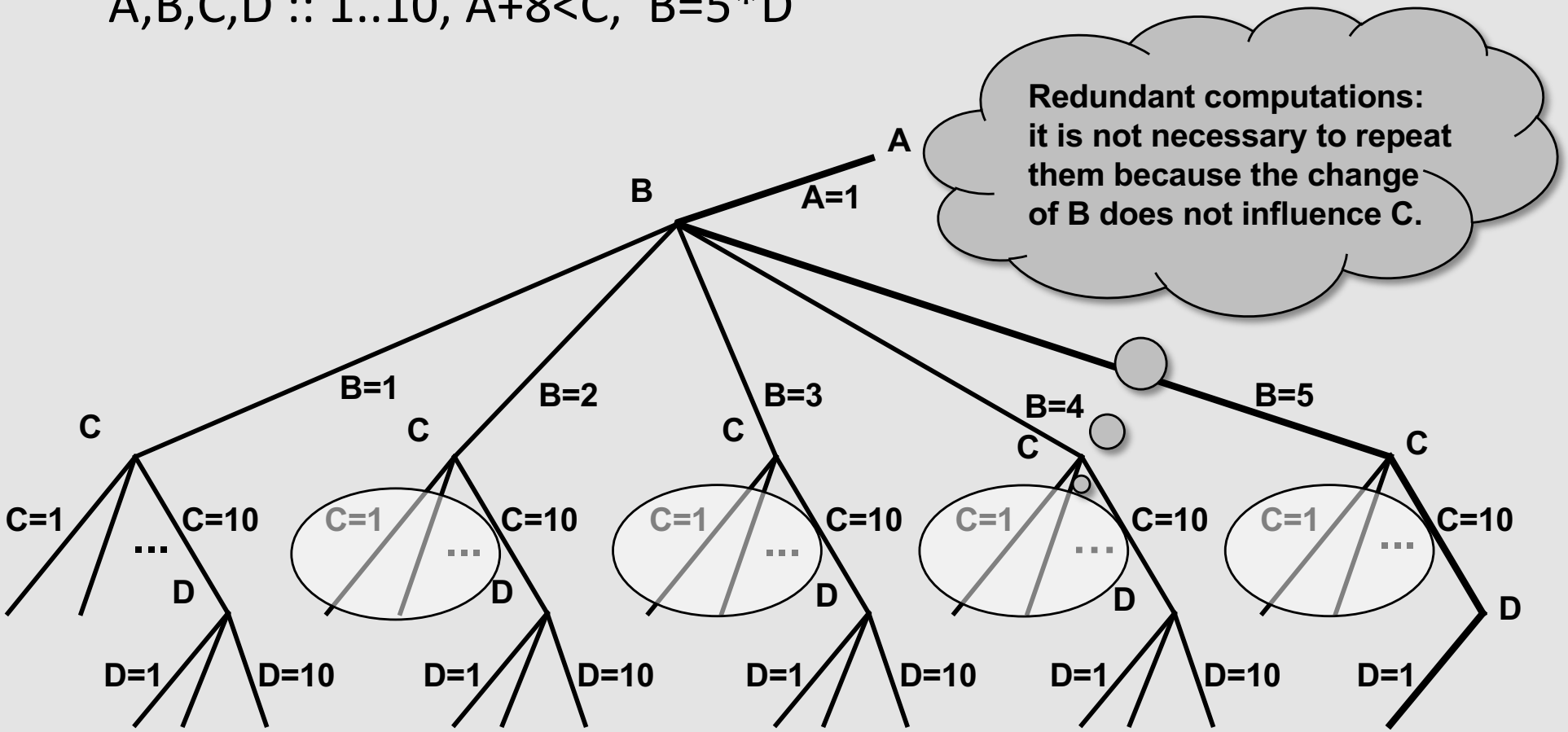


What is a redundant work?

– repeated computation whose result has already been obtained

Example:

A,B,C,D :: 1..10, A+8<C, B=5*D

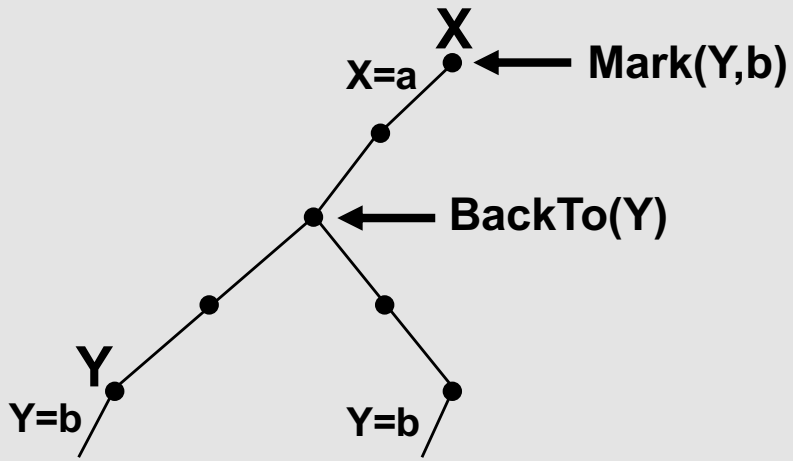


Remove redundant constraint checks by **memorising negative and positive results of tests**:

- **Mark(X,V)** is the farthest (instantiated) variable in conflict with the assignment X=V
- **BackTo(X)** is the farthest variable to which we backtracked since the last attempt to instantiate X

- Now, some constraint checks can be omitted:

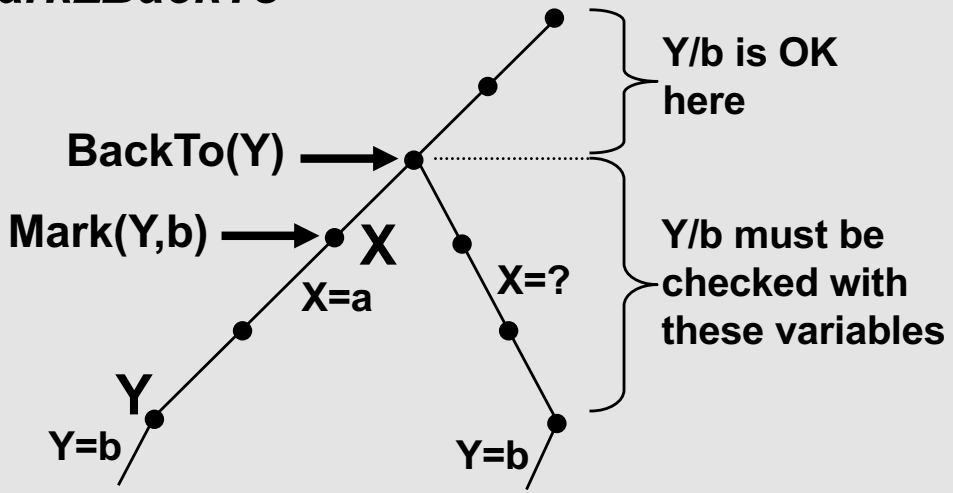
Mark<BackTo



Y/b is inconsistent with X/a (and consistent with all variables above X)

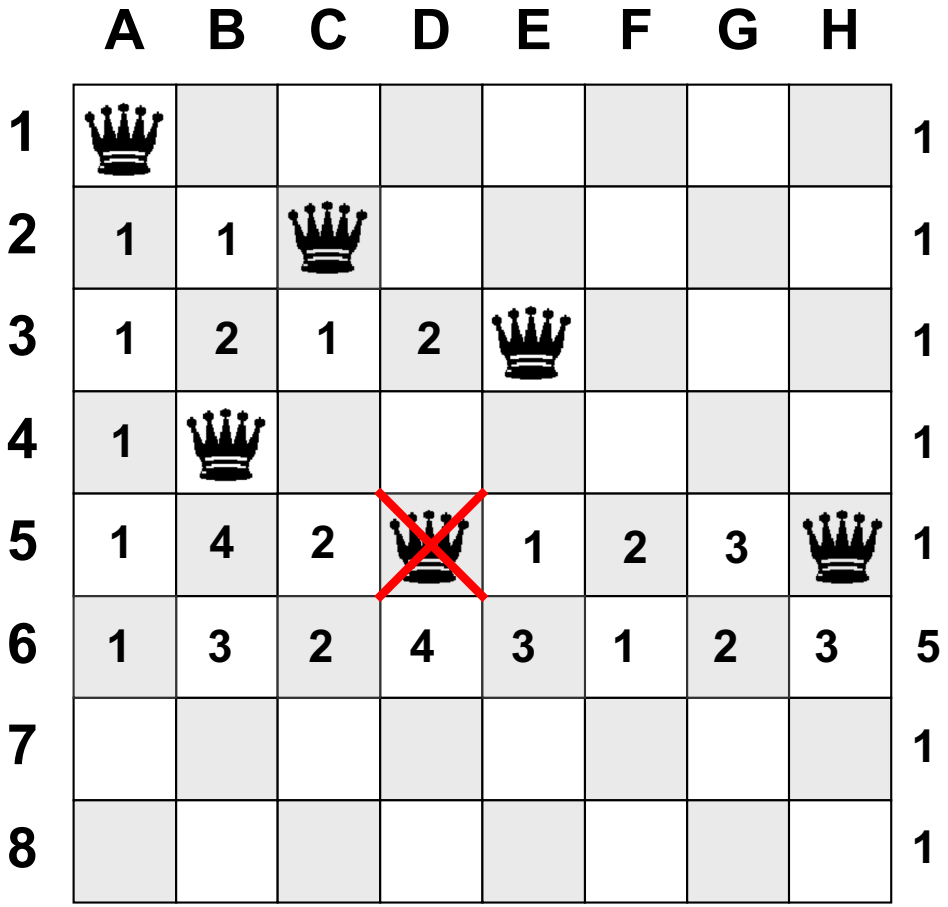
Y/b is still in conflict with X/a, we do not need to check it

Mark≥BackTo



Y/b is inconsistent with X/a (and consistent with all variables above X)

N-queens problem



1. Queens in rows are allocated to columns.

2. Latest choice level is written next to chessboard (BackTo). At beginning 1s.

3. Farthest conflict queen at each position (MarkTo). At beginning 1s.

4. 6th queen cannot be allocated!

5. Backtrack to 5, change BackTo.

6. When allocating 6th queen, all the positions are still wrong (MarkTo<BackTo).

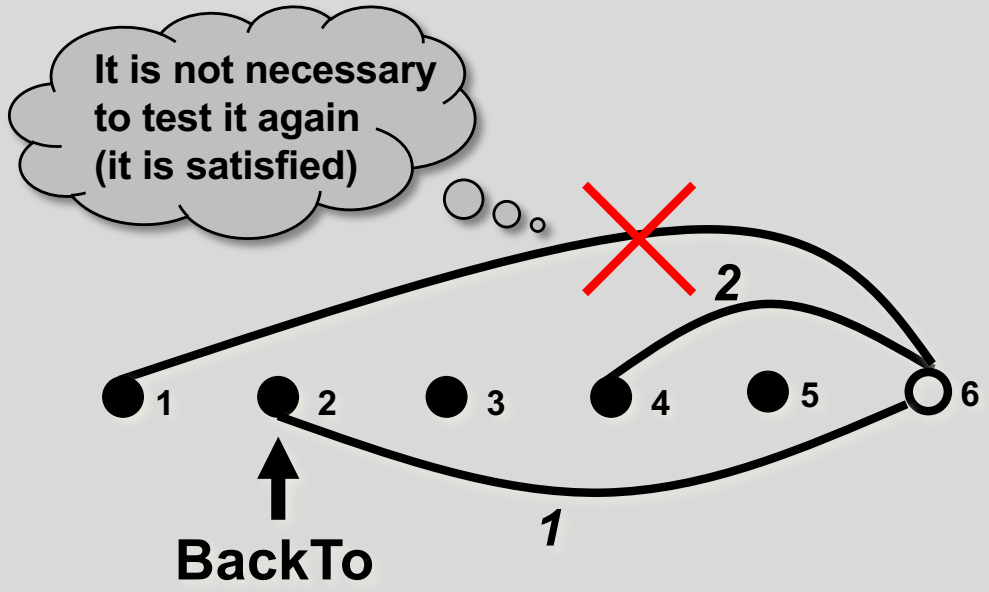
Note:

backmarking can be combined with backjumping (for free)

Consistency check for Backmarking

Only the constraints where any value is changed are re-checked, and the farthest conflicting level is computed.

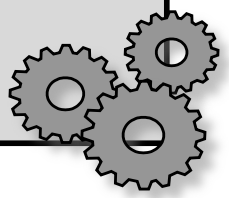
```
procedure consistent(X/V, Labelled, Constraints, Level)
    for each Y/VY/LY in Labelled such that LY≥BackTo(X) do
        % only possible changed variables Y are explored
        % in the increasing order of LY (first the oldest one)
        if X/V is not compatible with Y/VY using Constraints then
            Mark(X,V) ← LY
            return fail
        end if
    end for
    Mark(X,V) ← Level-1
    return true
end consistent
```

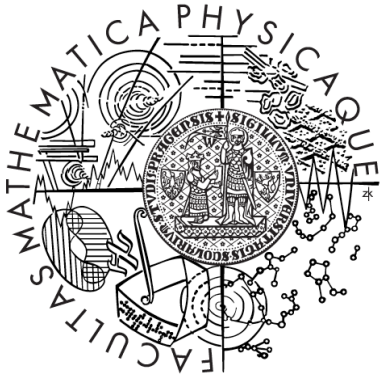


Algorithm Backmarking

rekurzivní verze

```
procedure BM(Unlabelled, Labelled, Constraints, Level)
  if Unlabelled = {} then return Labelled
  pick first X from Unlabelled          % fix order of variables
  for each value V from D_X do
     if Mark(X,V) ≥ BackTo(X) then      % re-check the value
        if consistent(X/V, Labelled, Constraints, Level) then
           R ← BM(Unlabelled-{X}, Labelled ∪ {X/V/Level}, Constraints, Level+1)
           if R ≠ fail then return R     % solution found
        end if
     end if
  end for
  BackTo(X) ←  Level-1                  % jump will be to the previous variable
  for each Y in Unlabelled do           % tell everyone about the jump
    BackTo(Y) ←  min {Level-1, BackTo(Y)}
  end for
  return fail                           % backtrack to the previous variable
end BM
```





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